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Paleomagnetic and petrologic evidence bearing on the age and origin of uranium deposits in the Permian Cutler Formation, Lisbon Valley, Utah

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ABSTRACT

An approximate age for uranium deposits in red beds of the Permian Cutler Formation, Lisbon Valley salt anticline, Utah, was obtained using paleomagnetic techniques. Progressive thermal demagnetization of samples of mineralized sandstone isolates stable magnetization components having high (≳400 °C) unblocking temperatures that define a tilt-corrected mean direction of D = 358.1°, I = 65.5°, α_{95} = 3.3°. This direction is close to expected Late Cretaceous to middle Tertiary mean directions at Lisbon Valley. In contrast, thermal cleaning of samples of unmineralized sandstone isolates tilt-corrected southeasterly, shallow mean directions (D = 140.5°, I = −9.5°, α_{95} = 8.6°, and D = 155.9°, I = −4.4°, α_{95} = 10.9°, for two localities in the Lisbon Valley area). These results are closely similar to results from the Cutler Formation elsewhere on the Colorado Plateau and from other Lower Permian strata in North America. In unmineralized sandstone, the stable remanent magnetization is carried predominantly by martite of postdepositional or detrital origin, whereas in mineralized sandstone, the stable magnetization is carried largely by authigenic specular hematite as uranium- and vanadium-bearing clusters in interstitial areas, and it thus reflects the time of mineralization. Partial dissolution of martite grains and grain-coating ferric oxide pigment prior to formation of abundant interstitial specular hematite has minimized or eliminated the influence of late Paleozoic magnetic components in mineralized sandstone.

These results, combined with information on the structural development of the Lisbon Valley area and on nearby uranium deposits in the Triassic Chinle Formation, suggest that the growth of the Lisbon Valley anticline during Late Cretaceous to early or middle Tertiary time promoted oxidative destruction of Chinle orebodies and led to a redistribution of uranium and iron from the Chinle into the Cutler. Acidic solutions generated during destruction of sulfide minerals in Chinle orebodies may account for the simultaneous transport of significant quantities of iron and uranium and for the partial dissolution of martite in the Cutler host beds. Neutralization of these solutions by reaction with host-rock constituents (especially calcite) caused precipitation of amorphous ferric oxide, which incorporated uranium by adsorption and which aged to form specular hematite.

INTRODUCTION

Known uranium ore deposits in the Lower Permian Cutler Formation are confined to Big Indian Valley on the southwest flank of the Lisbon Valley anticline, one of several northwest-trending, salt anticlines of the Paradox Basin (Figs. 1 and 2). These Cutler deposits differ fundamentally in ore mineralogy and in geochemical environment of ore formation from most other uranium deposits of the Colorado Plateau that are hosted by the Upper Triassic Chinle and the Upper Jurassic Morrison Formations. Many of the Chinle and Morrison deposits have been studied in detail during the past 30 years (for example, Page and others, 1956; Garrels and Larsen, 1959; Miller and Kulp, 1963; Fischer, 1970; Ludwig and others, 1984) and are well characterized with regard to ore mineralogy and mineralization ages. In contrast, the Cutler deposits have received only cursory examination, and their age has remained in question. In this article, we present results of a paleomagnetic study that was designed to obtain an approximate mineralization age of the Cutler ores. The paleomagnetic results, supplemented by petrologic studies, also constrain processes

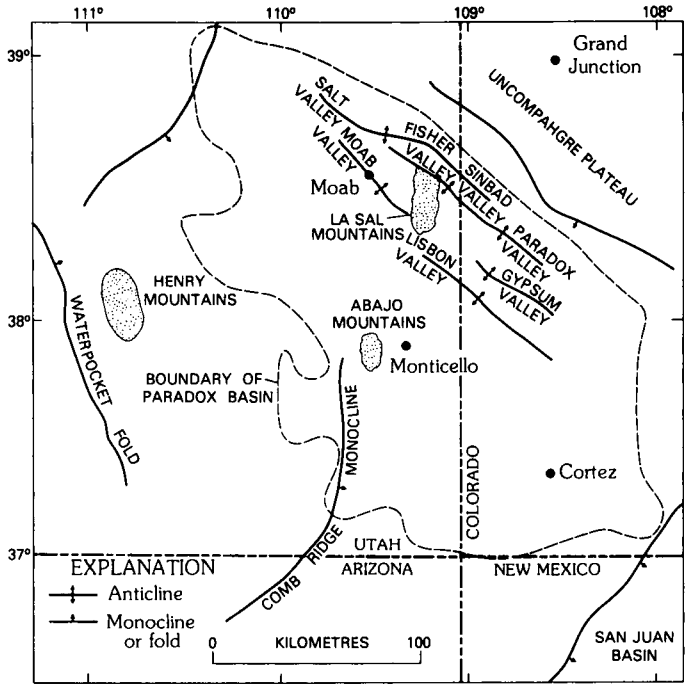


Figure 1. Map of part of the Colorado Plateau (modified from Butler and Fischer, 1978), showing location of the Lisbon Valley anticline.

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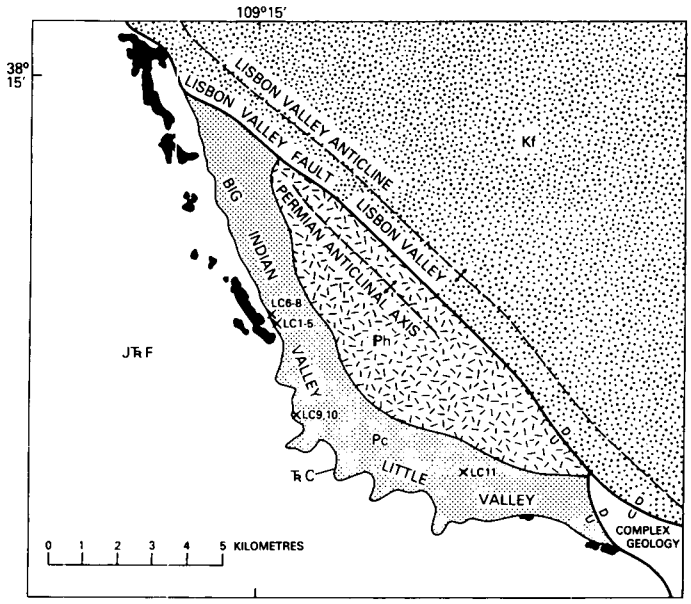


Figure 2. Simplified geologic map of the Big Indian Valley–Lisbon Valley area, Utah (modified from Butler and Fischer, 1978), showing sampling location (x) in the Cutler Formation and uranium deposits (black) at the base of the Chinle Formation, which compose the Big Indian mineral belt, projected vertically to the surface. Kf, mostly Cretaceous formations; JTr f, Jurassic and Triassic formations; Tr c, Triassic Chinle Formation; Pc, Permian Cutler Formation; Ph, Pennsylvanian Hermosa Formation; U, upthrown block; D, downthrown block.

responsible for introduction and concentration of uranium in the Cutler Formation.

The deposits of the Cutler Formation are not now, and never were, a major source of uranium. Their small production level, coupled with their limited distribution, largely explains the lack of attention paid to these deposits. Nevertheless, the Cutler deposits appear to represent an unusual and poorly understood type of stratiform orebody, and information on their age and origin presented in this paper extends understanding of the genesis of uranium deposits in general.

Uranium Deposits of the Colorado Plateau

The Cutler uranium deposits, which also contain large amounts of vanadium, occur as thin, elongate lenses or pods that are generally conformable with the boundaries of enclosing fluvial channel-fill arkosic sandstones (Weir and Puffett, 1981; Campbell and Steele-Mallory, 1979; Campbell, 1981). The channel sandstones are irregularly mottled in color by various shades of purple and red imparted by ferric oxide minerals and are locally bleached to light brown and white. Common uranium and vanadium minerals are carnotite, bequerelite, and a vanadium hydromica (Weir and Puffett, 1981), in which the uranium and vanadium are in oxidized valence states. The visible ore minerals typically occur as disseminated grains or as clusters of grains that are irregularly scattered throughout the host sandstone (Weir and Puffett, 1981). These minerals are also locally concentrated along some fractures and joints; such occurrences are uncommon and do not reflect primary controls on ore position, but, instead, appear to have formed by local uranium redistribution after the main ore-forming episode (Weir and Puffett, 1981). Campbell and Steele-Mallory (1979) investigated mineralized samples that lacked visible ore minerals, and, by examining fission-track maps, found substantial amounts of uranium to be coextensive with interstitial hematite. Other important authigenic host-rock constituents include copper minerals and calcite. The copper minerals are hydrous cupric carbonates and vanadium oxides and an unidentified copper-uranium mineral (Weir and Puffett, 1981). Calcite occurs as a massive cement in the sandstone remote from the ore zones but as sparse patches in the ores.

The Cutler host rocks contain little or no organic carbon, based on geochemical analyses, and they lack detrital organic matter and authigenic iron sulfide minerals or even iron oxide grains formed by oxidation of precursor sulfides (Campbell and Steele-Mallory, 1979; Campbell, 1981; Weir and Puffett, 1981). The Cutler deposits thus lack evidence of a present or former relationship to reducing conditions.

The major uranium deposits in the Chinle and Morrison Formations of the Colorado Plateau, in contrast, are found in chemically reduced environments. Common ore minerals in these deposits (uraninite, coffinite, and vanadium chlorite and oxide) form tabular or lenticular orebodies and are closely associated with organic matter, with iron disulfide (FeS_2-pyrite and marcasite) minerals, and, in the Chinle, with cuprous sulfide minerals such as bornite and chalcopyrite (Garrels and Larsen, 1959; Fischer, 1970). Some uranium deposits in the Morrison Formation are found in oxidized sandstone; ore minerals in these deposits, as in the Cutler, contain uranium and vanadium in oxidized valence states. Unlike the Cutler deposits, however, the oxidized Morrison ores are within the same beds as, and commonly adjacent to remnants of, unoxidized tabular (primary) orebodies (for example, Garrels and others, 1959; Elston and Botinelly, 1959). Dating of the tabular ores yields U^{235}-Pb^{207} isotopic ages---~210 m.y. for the Chinle (Miller and Kulp, 1963) and ~130 m.y. for the Morrison (Ludwig and others, 1984)—close to the ages of the respective host beds.

Stratigraphic and Structural Setting of the Cutler Deposits

The Cutler uranium deposits are limited in stratigraphic as well as in areal extent. All Cutler orebodies occur within 30 m of an angular unconformity separating the Cutler from the overlying Chinle. In the study area, the angular discordance is 4°. Two distinct episodes of uplift are responsible for the formation of the Lisbon Valley anticline and the present 14°–18° dips of Cutler beds on the southwest flank of the anticline. Initial tilting of Cutler sediments, which began in Permian time, was caused by upward squeezing of underlying Pennsylvanian evaporites (Butler and Fischer, 1978). Salt flowage, which formed the ancestral (late Paleozoic) Lisbon Valley anticline (Fig. 2), ceased by the end of Permian time and was followed by the erosional truncation of the anticline and the deposition of overlying Chinle sediments. Small variations in Chinle thickness and patterns of Chinle depositional facies suggest that salt flowage during the Triassic may have induced minor uplift of parts of the anticline, but the timing and amounts of this deformation are indeterminate (Butler and Fischer, 1978; Huber, 1980). In Late Cretaceous to early or middle Tertiary time, a major axial fault (~900 m of displacement) cut the late Paleozoic anticline, and the southwest anticlinal flank was tilted to its present attitude (Butler and Fischer, 1978). Fault displacement cannot account directly for the post-Permian tilting. Instead, fault movement may have induced salt flowage which caused beds to be upturned.

Dating the Cutler Uranium Deposits Using Paleomagnetic Methods

Geologic and petrographic studies have yielded vague and conflicting estimates for the age of the Cutler uranium deposits. The apparent geomet-

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ric conformity of the orebodies to the shape of the enclosing channel sandstones, which are noticeably more coarse-grained than surrounding barren sandstone, indicates a lithologic control on the position of the ore. This relationship suggests that the age of the uranium deposits is close to that of the host beds (Campbell and Steele-Mallory, 1979). On the other hand, diagenetic studies (also by Campbell and Steele-Mallory, 1979) provided evidence that interstitial uraniferous hematite formed relatively late in the sequence of alterations, an observation suggesting that mineralization occurred long after deposition of the Cutler.

Secondary enrichment of uranium minerals along fractures and joints, although uncommon, indicates that at least parts of the orebodies have been an open system with respect to gain or loss of uranium. Such uranium migration probably renders the Cutler ores unsuitable for isotopic age determination, and, to our knowledge, no such attempt has been made.

The petrologic observations of Campbell and Steele-Mallory suggested to us that uncertainties in the age of the Cutler deposits may be resolved using paleomagnetic methods. This approach was based on the premise that directions of remanent magnetization carried by authigenic uraniferous hematite might reflect to some extent the time of mineralization. If, for example, the Cutler ores formed relatively soon after deposition of the host beds, directions of remanent magnetization in the mineralized (ore-bearing) and unmineralized (barren) sandstones should be similar. If, however, authigenesis of uraniferous hematite occurred at a much later time with respect to apparent polar wander of North America, we anticipated finding remanent directions in mineralized sandstone unlike those in unmineralized sandstone remote from the orebodies. If this were the case, and if the precision of paleomagnetic data were sufficiently high, comparison of remanent directions in the ore sandstones with those derived from a polar wander path for North America would permit an estimate of the age of Cutler mineralization.

FIELD AND LABORATORY METHODS

Samples were collected at 11 sites in three widely separated areas in the Big Indian Valley–Lisbon Valley region (Fig. 2). Eight sites were located in the vicinity of the largest of the Cutler mines (Big Buck), five sites (LC1 to 5; 30 specimens) in unmineralized sandstone, and three (LC6 to 8; 44 specimens) in mineralized sandstones. Sites 1, 2, and 5 were located in dark reddish-brown (10R 3/4) sandstone at the stratigraphic level of the ore-bearing sandstone and at distances between 300–370 m south of the main mine adit (at LC7). Sites 3 and 4 were taken ~25 m above sites 1, 2, and 5. Sites 6–8 spanned a distance of ~90 m in grayish purple (5P 4/2) and pale red-purple (5RP 6/2), ore-bearing sandstone. Ore sandstone, similar to that at sites 6–8 in grain size and color, was also sampled in the Spiller Canyon area (LC9, LC10; 19 specimens) 3 km south of the Big Buck adit. A final site (LC11; 13 specimens) in barren sandstone (10R 3/4) was collected in Little Valley; this site occurs in nearly the same stratigraphic position as that represented by sites 9 and 10, based on its relative position to an aeolian sandstone marker bed that has been traced to Spiller Canyon. All sites lie within 30 m of the overlying Chinle. Fossil evidence and regional geologic and stratigraphic relationships (Baars, 1962; McKee and others, 1967; J. A. Campbell, unpub. data) indicate that the sampled units are in the lowermost Cutler (Wolfcampian in age). The absence of the upper Cutler is due either to nondeposition or to erosion caused by uplift of the anticline prior to Chinle deposition.

At each site, samples were drilled at the outcrop using a portable gasoline-powered drill and oriented with a magnetic compass. Samples at sites 7 and 8 also were oriented using a sun compass because of the presence of metal in the vicinity of the adits; corrections of magnetic declinations were not required. Directions and intensity of remanent magnetization were measured using 5-Hz spinner magnetometer with a sensitivity of $\sim 10^{-4}$ A m^{-1} (10^{-7} emu cm^{-3}). Forty-four specimens were subjected to progressive alternating field (AF) demagnetization in a tumbling degausser, typically in 10 to 12 peak-field steps up to 100 mT. Sixty-seven specimens were thermally demagnetized in a noninductively wound furnace in which the cooling field, attenuated by nested mu-metal cylinders, is ±4 nT. Heating was undertaken in steps of 100 °C up to 600 °C, and thereafter in increments of 15 to 50 °C up to 680 or 700 °C. All demagnetization results were plotted on orthogonal vector (demagnetization) diagrams (Zijderveld, 1967); components of magnetization represented by linear, or nearly linear, segments containing three or more points were calculated by vector subtraction.

We considered the verification of uraniferous hematite, strongly implied by fission-track maps (Campbell and Steele-Mallory, 1979), to be crucial in establishing a relationship between the age of mineralization and the remanence carried by ore-zone hematite. For this purpose, we used a scanning electron microscope (SEM) with an attached energy-dispersion X-ray analyzer (EDA). This method also provided information on differ-

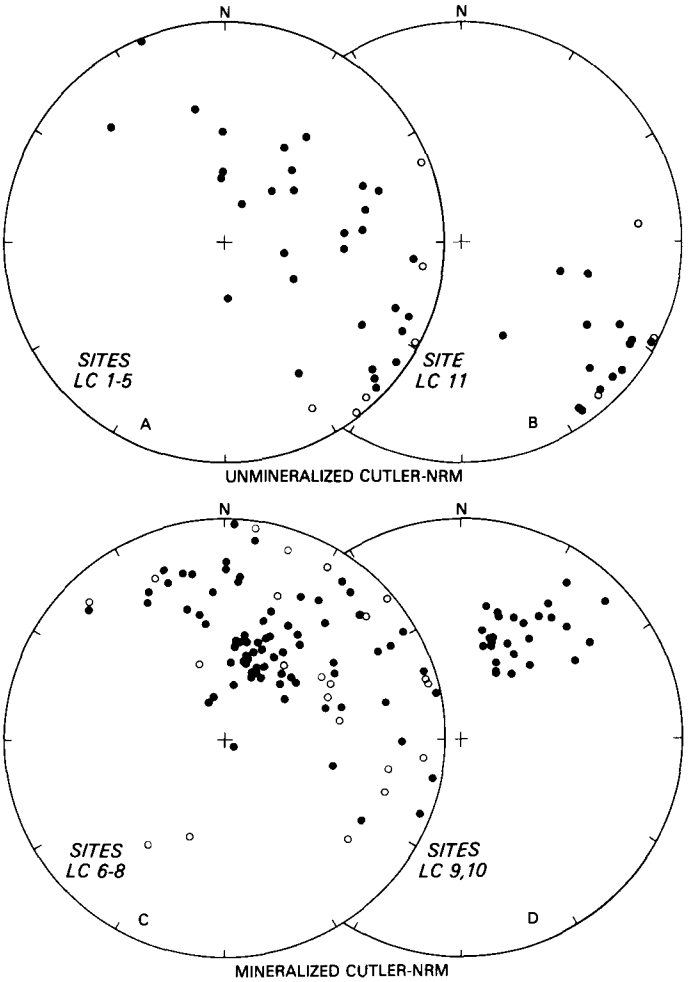


Figure 3. Equal-area stereographic projection of directions of natural remanent magnetization (NRM) of unmineralized (A and B) and mineralized (C and D) sandstones of the Cutler Formation, Lisbon Valley area, Utah, without tilt correction. In these and following stereographic plots (Figs. 4, 6, 8, 10), solid (open) symbols represent lower (upper) hemisphere projections. A, sites LC1–LC5; B, site LC11; C, sites LC6–LC8; D, sites LC9 and LC10.

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ences in shape and size between hematite in the ore zones and that in the barren sandstones. For SEM-EDA analysis, we examined nine fragments from ore-zone and two from barren-rock specimens. SEM-EDA analysis was supplemented by study of polished thin sections (five from barren and eight from ore sandstone) that were viewed both in transmitted light and under oil in vertically reflected light.

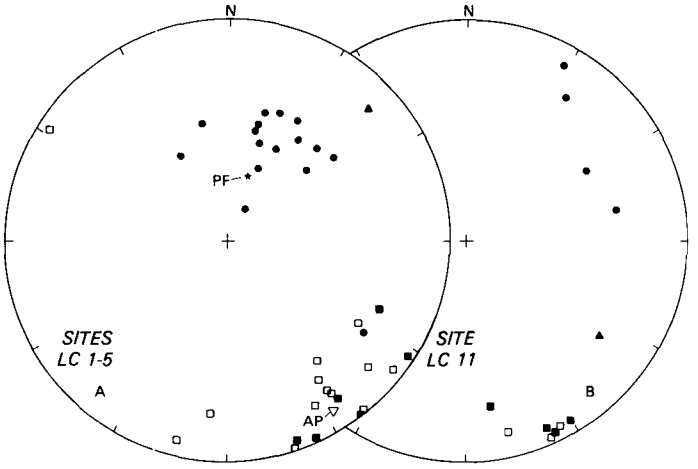


Figure 4. Equal-area stereographic projection of components of magnetization (without tilt correction) isolated by progressive thermal demagnetization of samples of unmineralized sandstone, Cutler Formation. Circles represent low-T_{ub} components; small solid triangles, intermediate-T_{ub} components; and squares, high-T_{ub} components (see text). A, sites LC1–LC5; B, site LC11. Present field (PF) and average Early Permian (AP, open inverted triangle; Diehl and Shive, 1979) directions are shown in A.

RESULTS

Paleomagnetic Data

Directions of natural remanent magnetization (NRM) of specimens from most sites are broadly scattered. Nearly all NRM directions of barren sandstone from sites LC1-5 (Fig. 3A) are directed either to the north and northeast with positive inclinations, or to the southeast with shallow positive and negative inclinations. This latter direction is the dominant NRM direction of specimens from site LC11 (Fig. 3B). Directions of NRM in ore-bearing sandstone (sites LC6–10; Figs. 3C and 3D) are also widely scattered but are most densely concentrated in the northeast quadrant. Average NRM intensities are lowest in barren sandstone at sites LC1-5 (2.9×10^{-3} A m^{-1}) and highest in mineralized rock at sites LC9 and LC10 (9.5×10^{-3} A m^{-1}). AF cleaning causes little or no directional change in most specimens and thereby fails to reduce appreciably the scatter of NRM directions. Such behavior, indicative of high coercivity of magnetic components, is common to many red-bed units.

Thermal demagnetization reveals four components of magnetization—two dominant paleofield directions, and, in a few specimens, their antipodal equivalents. These directions are carried by magnetic minerals with broad and partly overlapping unblocking temperature (T_{ub}) spectra, so the ranges of demagnetizing temperatures, in which the different directions are removed, vary among different specimens. Most thermally cleaned specimens (~80%) contained at least two remanent components; one component with low T_{ub} was typically removed at temperatures below 300° or 400 °C, and the other, with high T_{ub}, was usually defined after treatment above 400° to 600 °C. Nearly 10% of the specimens contained only one remanent component; in the following discussion, these are considered as high-T_{ub} components.

Vector demagnetization paths of the remaining specimens have, in addition to low- and high-T_{ub} components, nearly straight decay segments over a range of intermediate temperatures (~300°–600 °C), and these

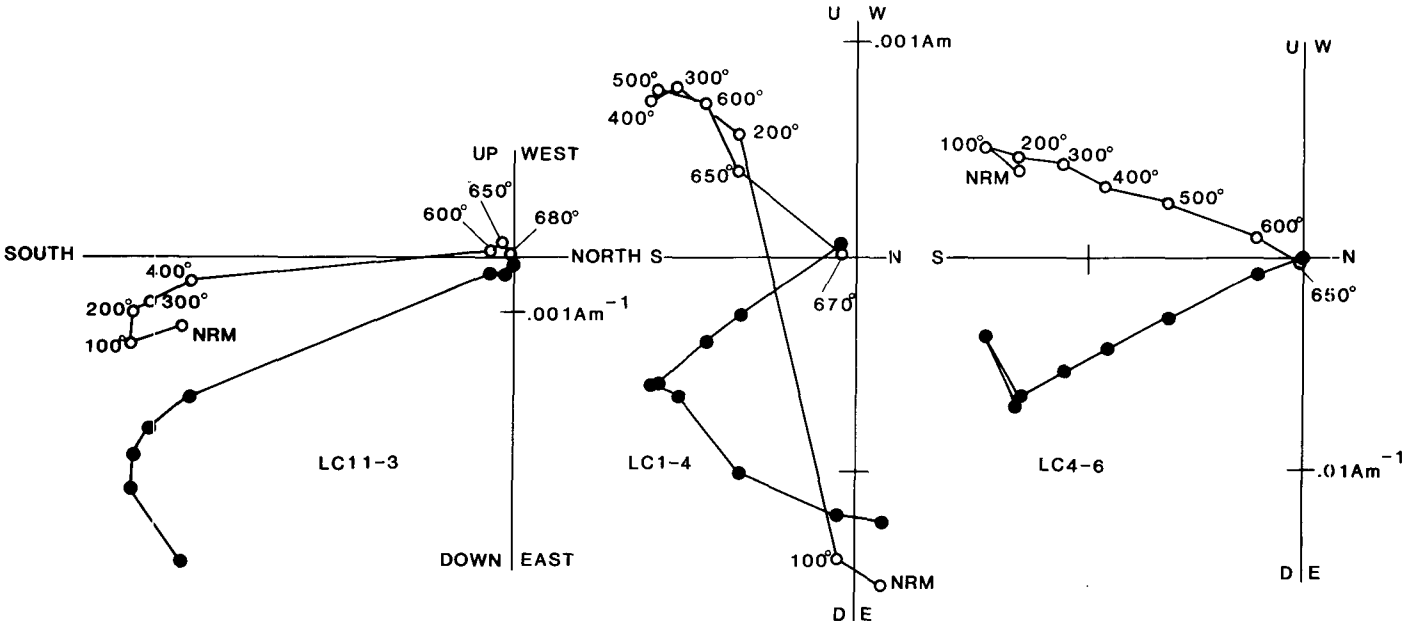


Figure 5. Orthogonal progressive demagnetization projections (Zijderveld, 1967) of the successive endpoints of the remanence vector during thermal demagnetization, temperature in degrees centigrade, of samples from unmineralized sandstone. Projections of the endpoints onto the vertical (horizontal) plane are indicated by open (closed) symbols and are plotted without correction for tilt of the strata (in situ coordinates).

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TABLE 1. SUMMARY OF PALEOMAGNETIC DATA: CUTLER FORMATION, LISBON VALLEY, UTAH

Site	Latitude (°N)	Longitude (°E)	N	D	I	α_{95}	k	R	D_c	I_c	Virtual geomagnetic pole Latitude (°N)	Virtual geomagnetic pole Longitude (°E)	δp	δm
Unmineralized														
LC1, 2, 3, 4, 5	38.20	250.75	16	143.0	-10.3	8.6	19.2	15.2202	140.5	-9.5	41.0	127.9	4.4	8.7
LC11	38.15	250.81	7	158.6	3.5	10.9	31.8	6.8113	155.9	-4.4	47.8	108.2	5.5	10.9
Mineralized														
LC6	38.20	250.75	4	19.2	62.5	12.7	52.9	3.9433	3.8	69.9	74.2	259.0	18.7	21.8
LC7	38.20	250.75	7	7.1	63.0	3.8	251.2	6.9761	348.4	68.6	74.1	223.9	5.4	6.4
LC8	38.20	250.75	9	5.2	59.9	7.6	46.3	8.8272	348.9	65.4	77.6	213.3	10.0	12.3
LC9	38.17	250.76	8	16.2	53.3	5.8	92.9	7.9246	5.6	60.6	84.5	300.5	6.7	8.8
LC10	38.17	250.76	4	20.6	55.1	14.2	42.6	3.9796	10.1	62.9	80.2	298.4	17.5	22.3
LC6–LC10 (specimens)	38.19	250.75	32	12.4	58.8	3.3	59.4	31.4781	358.1	65.5	80.4	243.0	4.3	5.4
LC6–LC10 (sites)	38.19	250.75	5	14.0	58.9	5.4	204.4	4.9804	359.9	65.8	80.1	250.4	7.2	8.8

Note: data represent high-T_{ub} components determined by vector subtraction (see text).

N, number of specimens; D, *in-situ* declination in degrees from true north; I, *in-situ* inclination in degrees, positive downward; α_{95}, semi-angle of the cone of 95% confidence in degrees; k, precision parameter (Fisher, 1953); R, resultant vector; D_c, tilt-corrected declination in degrees; I_c, tilt-corrected inclination in degrees; δp and δm, semi-minor and semi-major axes (in degrees), respectively, of the oval of 95% confidence surrounding the virtual geomagnetic pole.

paths thus suggest the presence of an additional component. For most specimens that exhibit this behavior, the apparent intermediate-T_{ub} components are probably vector resultants that reflect nearly constant contributions from two different components over these temperatures. In a few specimens, however, intermediate-T_{ub} components appear to represent ancient field directions, and these directions may thus provide useful information on the magnetization history of the Cutler sandstones.

Thermal cleaning of unmineralized specimens from sites LC1–5 reveals two components of magnetization: one component, with low T_{ub} and found in most specimens, is defined by removed vectors having north-easterly declinations and positive inclinations, and the other, with high T_{ub}, has southeasterly declination and shallow positive and negative inclination (Figs. 4A and 5). In one specimen, the direction removed at high temperatures was nearly opposite to this latter direction. The mean direction of the low-T_{ub} component, without correcting for bedding tilt (in situ) and omitting one direction in the southeast quadrant (Fig. 4A), is D = 21.5°; I = 50.4° (α_{95} = 8.2°). This direction is close to that of the Earth's present field direction (D = 14°, I = 66°) at the site. Elimination of this component at high temperatures isolates a magnetic component with a tilt-corrected mean direction of D = 140.5°; I = -9.5° (Table 1). This direction was calculated by including the northwest, shallow direction inverted 180°, after omitting three specimens, two having south-southwest declinations (Fig. 4A) and one with an uninterpretable demagnetization path.

A similar result is obtained for barren sandstone at site LC11, except that directions close to the present field direction are absent (Figs. 4B and 5). Directions removed at low temperatures tend to be poorly defined as shallow in the northeast and may be resultants of two or more superimposed magnetizations. Higher demagnetizing temperatures are effective in revealing southeast, shallow directions which give a tilt-corrected mean direction of D = 155.9°, I = -4.4° (Table 1).

Specimens from sites in mineralized sandstone collectively exhibit similar magnetic behavior in response to thermal treatment (Figs. 6–8). Most directions removed at low and intermediate temperatures lie in the northeast quadrant. Interestingly, three specimens have northwesterly, shallow directions that when inverted 180° give an in situ mean direction (D = 145.4°, I = -7.0°, α_{95} = 11.3°) similar to the mean directions determined from high-T_{ub} components in barren sandstone. Moreover, continued cleaning of one of these specimens (LC8-6) removed a direction nearly opposite that of its low-T_{ub} component (Fig. 6A), indicating that the specimen's remanence was acquired during periods of opposite polarity.

Demagnetization above ~400°–600 °C isolates directions with north-northeast declinations and steep positive inclinations (Figs. 7 and 8). In three specimens, an opposite direction is removed, and in another, the southeast, shallow direction is found (Fig. 8A). The remanent directions shown in Figure 8 represent the stable, characteristic (see Zijderveld, 1967) components of magnetization in the mineralized sandstone after attempts to remove all spurious influences by other magnetizations have been made. The apparent elongation (to the northeast with decreasing inclination) seen in the north-northeast group of directions (Fig. 8B), however, implies that not all such influences have been entirely eliminated from some specimens. Nevertheless, because of an insufficient basis on which to reject any of these directions, all are included in the mean direction of mineralized Cutler sandstone. After inverting the three southerly, negative directions, the in situ mean direction is D = 12.4°, I = 58.8° (Table 1). The specimen having the southeast, shallow direction and five others that showed unstable response during demagnetization were not included in this calculation. An alternate representation of the mean direction is derived by averaging individual site-mean directions. Using data for

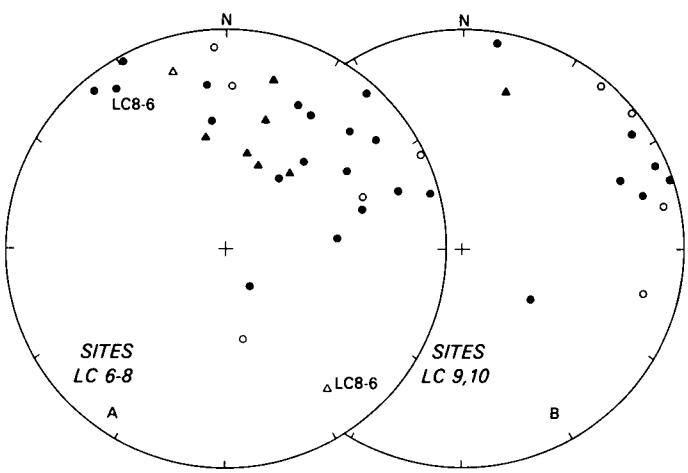


Figure 6. Equal-area stereographic projection of low- and intermediate-T_{ub} components of magnetization (without tilt correction) isolated by progressive thermal demagnetization of samples from mineralized sandstone, Cutler Formation. Symbols as in Figure 4. A, sites LC6–LC8; B, sites LC9 and LC10.

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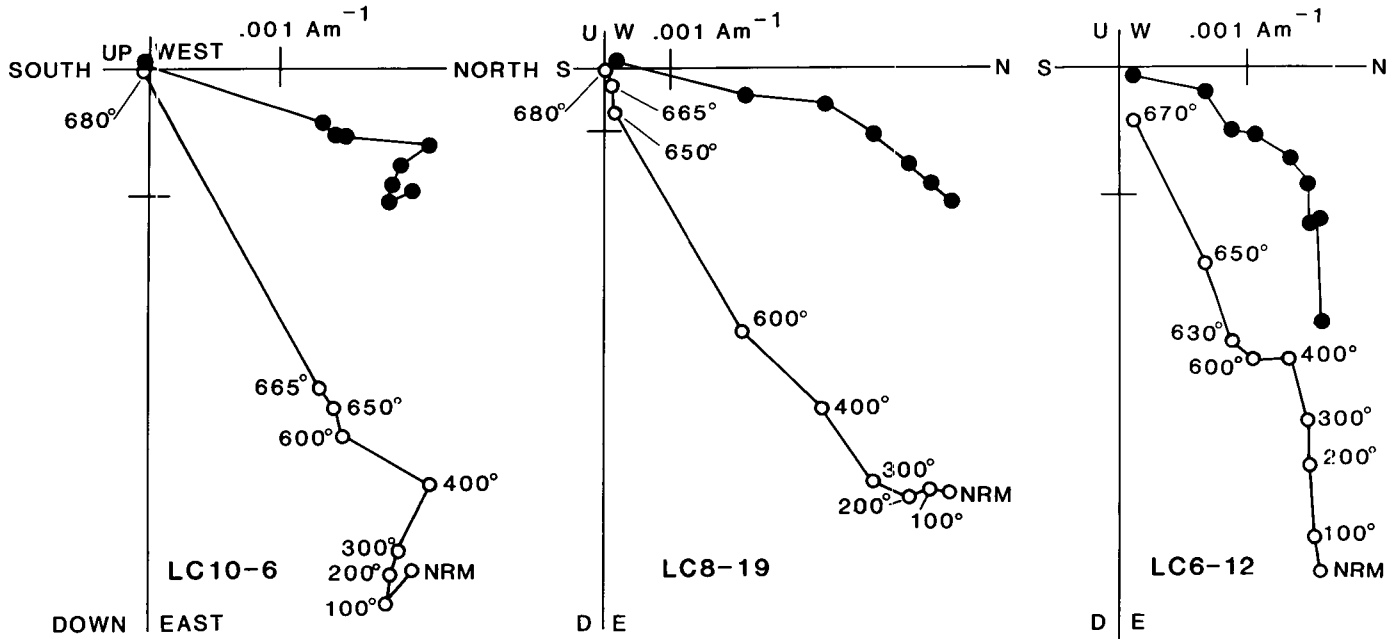


Figure 7. Orthogonal progressive demagnetization projections of the successive endpoints of the remanence vectors during thermal demagnetization, temperature in degrees C, of samples from mineralized sandstone. Data are plotted with respect to in situ coordinates. Symbols as in Figure 5.

sites LC6–10, the unit mean direction (in situ) is D = 14.0°, I = 58.9° (Table 1). The angular difference between these directions is <1°.

Petrologic Observations

Petrologic study focused on identifying magnetic minerals in both unmineralized and mineralized rock and on examining the texture and composition of ferric oxide minerals in the ore zones. Hematite, in a variety of forms and occurrences, is the dominant magnetic oxide mineral in the Cutler Formation. Two types of hematite—very fine-grained pigment and specular hematite—were seen in unmineralized sandstone. The fine-grained variety, which in polished thin sections appears as a bright red or red-orange translucent phase, occurs as coatings (<2 μm thick), primarily around detrital grains (Fig. 9A) and along cleavage planes within altered biotite. Even at magnifications as much as 35,000×, some of this pigment lacks discernible crystal form and thus may be amorphous or finely crystalline with individual crystals <0.1 μm in any dimension (see Walker and others, 1981). Other areas of grain-coating ferric oxide have the appearance of coalesced beads (each ~1 to 3 μm across) having poorly defined crystal faces. Most pigmentary ferric oxide directly coats surfaces of detrital grains and is covered commonly by authigenic clay, which, based on morphology and qualitative chemical analysis by EDA, is probably a smectite. Specular hematite is present primarily as rounded grains (~20–120 μm across) of martite (polycrystalline specularite that has pseudomorphically replaced pre-existing magnetite; Fig. 9B) and rarely as patches lacking crystallographic orientation within partly altered ilmenite grains. A few martite grains in samples from site LC11 contain sparse relicts of magnetite (≤6 μm across); the paucity of these relicts and results of thermal demagnetization indicate that magnetite does not noticeably contribute to remanence. Firm textural evidence bearing on the origin of the martite, whether detrital or postdepositional, is lacking. Nevertheless, these grains have remained essentially unaltered since they were hematitized, and their original margins are clearly defined. Specularite was also found in very small (<10 μm long) pods within a few biotite grains in specimens from site LC11. More commonly, biotite grains at this and other sites lacked specular hematite.

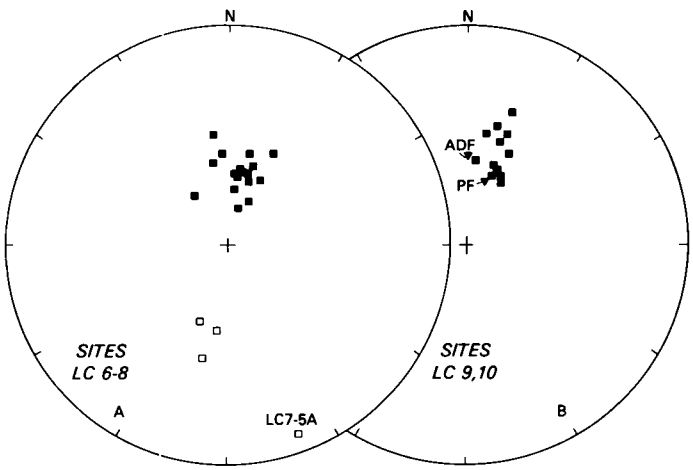


Figure 8. Equal-area stereographic projection of high-T_{ub} components of magnetization (without tilt correction) isolated by progressive thermal demagnetization of samples from mineralized sandstone, Cutler Formation. Symbols as in Figure 4. A, sites LC6–LC8; B, sites LC9 and LC10. Axial dipole field (ADF) and present field (PF) directions are shown in B.

Hematite is the only magnetic oxide mineral found in mineralized sandstone where it occurs also as fine-grained varieties and as specularite. Three fundamental differences, however, exist between hematite in the ore zones and hematite remote from these zones. The first is the presence in mineralized sandstone of specular hematite as nearly spherical clusters (~5 to 20 μm across) that form an interstitial cement, fill secondary voids left by dissolution, and coat detrital grains. The occurrences of this specularite indicate an authigenic origin. Viewed with the SEM, the clusters are seen to consist of thin and tightly interlocking euhedral blades (Fig. 9C), and in this regard, the clusters resemble authigenic hematite rosettes observed in

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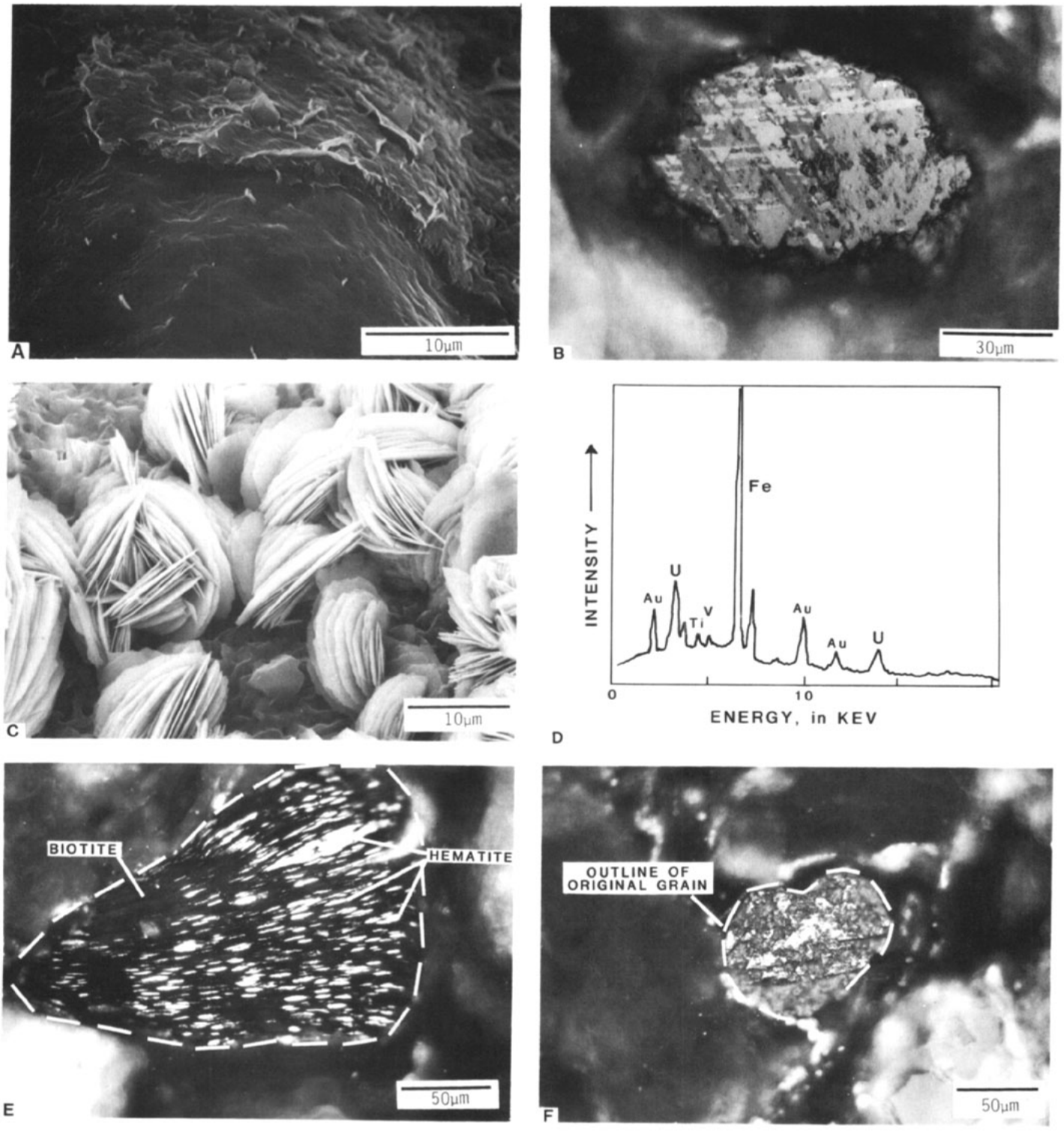


Figure 9. A. SEM photomicrograph of unmineralized sandstone (site LC11) showing ferric oxide as thin coating on detrital quartz grain. The ferric oxide appears to be intergrown with, and is partly overlain by, sparse clay (smectite?). B. Polished surface photomicrograph (under crossed nicols) of a polycrystalline hematite (martite) grain. The trellis pattern results from different orientations of crystalline units that intersect along (111) planes inherited from original magnetite grain. Unmineralized sandstone, site LC11. C. SEM photomicrograph of clusters of specular hematite composed of interlocking euhedral hematite blades. The clusters lie atop and thus postdate smectite(?). Mineralized sandstone, site LC10. D. Energy-dispersive X-ray spectrum for SEM probe beam directed on a blade within a specular hematite cluster from the ore zone in the Spiller Canyon mine. The hematite is enriched in uranium (U) and minor amounts of vanadium (V) and titanium (Ti). Sample coated with gold (Au). E. Polished surface photomicrograph of altered biotite grain that has been partly replaced by specular hematite and that contains specular hematite between cleavage lamellae. Mineralized sandstone, site LC7. F. Polished surface photomicrograph of altered polycrystalline hematite (martite). Fine-grained translucent red hematite now occurs in areas formerly occupied by specular hematite. Mineralized sandstone, site LC7.

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other ancient red beds (Walker and others, 1981). The hematite clusters in mineralized sandstone lie on top of, and hence formed after, authigenic clays that by shape and composition are identical to clays which postdate the grain-coating ferric oxide pigment in barren sandstone. Paragenetic relationships show that the clusters also postdate authigenic quartz crystals.

SEM-EDA examination confirms the intimate association of hematite and uranium suspected from fission-track maps. Probe beams (<1 μm diam) reveal a substantial quantity of uranium (estimated to be several thousand ppm, based on experience with well-characterized uraniferous materials), along with minor amounts of vanadium in hematite clusters from the Spiller Canyon mine (Fig. 9D). At magnifications of nearly 40,000× (giving adequate visual resolution to <0.1 μm), no discrete uranium minerals were seen on the surfaces of hematite blades, suggesting that the uranium occurs within the crystal structure of the hematite. Although uranium was not detected in hematite clusters in samples from outcrops or the other mine areas, minor amounts of vanadium in these clusters were commonly found. Titanium is also a minor but common element in the clusters in each sample examined.

Another major difference between ore-zone and barren-rock hematites is that in ore-zone samples, specular hematite is ubiquitous within detrital biotite grains (Fig. 9E). In this association, the hematite occurs as pods and as continuous thin bands (as much as 200 μm long) along cleavage lamellae. In biotite grains that have been bent and folded (in some cases through angles >90°) by compaction between framework silicate grains, unbroken specularite bands follow the cleavage folds and thus suggest introduction of ferric oxide after compaction.

The third disparity in ferric oxide forms is the paucity of very fine-grained pigment and the apparent alteration of polycrystalline specular hematite grains (predominantly martite) in mineralized sandstone. In the ore zones, red translucent pigment as thin borders on detrital grains is extremely rare, but it has been recognized beneath pore-filling calcite and in a few places beneath hematite clusters. Most ore-zone martite grains consist of small patches of specular hematite within red translucent hematite that defines the original grain margins (Fig. 9F); some of these grains are surrounded by clusters of authigenic specular hematite. This occurrence contrasts greatly with martite grains in barren sandstone that have remained largely unmodified since hematite replaced magnetite. These observations indicate that much of the ore-zone pigment and martite was postdepositionally destroyed before formation of the delicate hematite clusters.

Calcite in mineralized sandstone also differs from that in barren sandstone. Remote from ore, calcite occurs as massive, pore-filling cement making up >10 volume percent of the whole rock (Campbell and Steele-Mallory, 1979; this study). In contrast, calcite is sparse (4% and less) in mineralized rock (Campbell and Steele-Mallory, 1979; this study); where present, calcite forms irregular and isolated, but optically continuous, patches. These observations point to substantial dissolution of originally more abundant calcite cement in the ore zones.

DISCUSSION

Origin of Remanent Magnetization

Two dominant directions—one having north-northeast declination and steep positive inclination and the other having southeast declination and shallow, mostly negative inclination—are found in specimens from most sites of both mineralized and unmineralized sandstone. Thermal cleaning of unmineralized sandstone, however, removes the north-northeast component having low T_{ub} and isolates the southeast component having high T_{ub}, whereas heating of mineralized sandstone to high temperatures isolates the north-northeast direction (or, in a few specimens, the opposite direction). This dichotomy results from differences in the nature and occurrences of magnetic carriers in the mineralized and unmineralized sandstones.

The removal of directions, which are aligned close to the present Earth's field, by low-temperature demagnetization of unmineralized rock (sites LC1–5; Fig. 4A) is common behavior for many ancient red beds (Gose and Helsley, 1972; Diehl and Shive, 1979; Reeve and Helsley, 1972; Purucker and others, 1980). Such magnetically soft (with respect to temperature) behavior is difficult to trace directly to a magnetic phase, but, based on studies by Collinson (1974), it is generally attributed to ferric oxide pigment. The pigment may be composed of very fine-grained hematite and (or) unordered or poorly crystallized precursors to hematite (see Walker and others, 1981).

The origin of the low- and intermediate-T_{ub} components with shallow inclination that constitute a poorly defined belt in the northeast quadrant are even more problematic. These directions may result from nearly equal contributions from both shallow southeast and northwest remanent

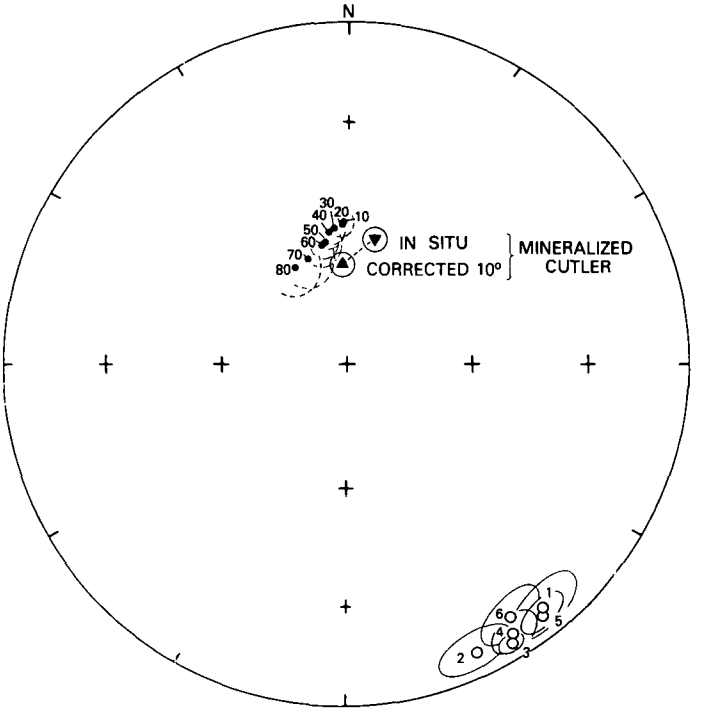


Figure 10. Equal-area stereographic projection of tilt-corrected mean remanent directions surrounded by α_{95} confidence cones (Table 1), Cutler Formation, Lisbon Valley area, Utah. Directions of unmineralized sandstones (point 1 denotes sites LC1–LC5; point 2, site LC11) plot in the southeast quadrant. Mean directions of the Cutler Formation elsewhere on the Colorado Plateau and of an average Early Permian direction (all corrected for Lisbon Valley geographic coordinates) are shown for comparison: points 3–5 denote Cutler directions (Gose and Helsley, 1972; Helsley, 1971); point 6, average Early Permian direction (Diehl and Shive, 1979). The in situ and tilt-corrected mean directions (triangles) of mineralized sandstone (Table 1) are compared with average directions derived from an apparent pole wander path for North America (Irving and Irving, 1982) and recalculated for Lisbon Valley geographic coordinates. Numbers adjacent to average North American directions refer to their approximate ages in 10^6 yr (see Irving and Irving, 1982).

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directions (see Larson and Walker, 1982), with an additional influence from normal (recent field) overprint. Strong evidence for this relationship is lacking, although one specimen (site LC1) does possess a shallow northwest direction (Fig. 4A).

Based on magnetic response to high demagnetizing temperatures (see Collinson, 1974) and our petrographic study, the abundant grains of specular hematite are thought to carry the stable remanence (high-T_{ub} components) in barren sandstone. The dominant form of specular hematite, martite, may have formed before or after deposition. If the martite is detrital, the stable remanence is then largely a depositional remanent magnetization (DRM), but if martite is authigenic, the stable remanence is most likely a chemical remanent magnetization (CRM). In this study, the distinction between DRM and CRM is not crucial. Nevertheless, the presence of sparse relict magnetite within martite in specimens from LC11 indicates that magnetite was a detrital constituent and implies that it was originally more abundant than at present because of its long-term exposure to oxidizing conditions. Moreover, the northwest shallow component isolated in one sample at a site from which other samples possessed the opposite direction is consistent with CRM acquisition over a sufficiently long time to span a field reversal. Regardless of the origin of the stable remanence, mean directions of unmineralized sandstone lie close to remanent directions of the Cutler elsewhere on the Colorado Plateau and of other Lower Permian strata in western North America (Helsley, 1971; Gose and Helsley, 1972; Diehl and Shive, 1979; Fig. 10).

Thermal demagnetization of mineralized sandstone yields some insight into the magnetizations of unmineralized specimens. Directions removed at low and intermediate temperatures from a few ore-zone specimens are nearly opposite to the reversed direction of unmineralized sandstone (Fig. 6A). Moreover, the behavior of specimen LC8-6 (Fig. 6A), having an intermediate-T_{ub} component nearly antiparallel to its low-T_{ub} component, suggests that this rock acquired its magnetization over a time interval during which the Earth's field reversed its polarity. These observations strengthen the inference that the shallow, mainly northeasterly directions removed from both ore and barren sandstone result partly from reversed and normal late Paleozoic and possibly early Mesozoic components.

Remanent directions in mineralized sandstone that are isolated at high demagnetizing temperatures cluster closely in a group far distant from the late Paleozoic directions defined for unmineralized rocks (Figs. 4 and 8). Again, magnetic behavior and petrologic studies indicate that specular hematite, in this instance as authigenic clusters and as pods and bands within biotite, is responsible for the stable magnetizations of ore-zone specimens. Remnants of severely altered specularite grains (predominantly martite) and perhaps sparse relicts of grain-coating ferric oxide pigment presumably still retain the late Paleozoic direction in these specimens, but, except in one (LC7-5A; Fig. 8A), the contribution of this hematite to total remanence is minimal or is totally overwhelmed because of the dominance of younger authigenic specularite. Specular hematite authigenesis in mineralized sandstone must have spanned at least one field reversal, because in three specimens, a direction opposite the north-northeast direction is isolated (Fig. 8A).

Age of Mineralization

The mean remanent direction of mineralized rock is considered to reflect closely the time of uranium mineralization, because stable remanence is carried predominantly by authigenic uranium- and vanadium-bearing specular hematite. This direction records a much younger paleomagnetic field than do directions of unmineralized rock. The time of uranium mineralization can be estimated by comparing the mean direction of mineralized Cutler sandstone with mean directions derived from an apparent polar wander path for North America (Irving and Irving, 1982) and recalculated for Lisbon Valley geographic coordinates (Fig. 10). The mean directions for North America considered in this comparison represent middle Cretaceous and younger paleomagnetic fields because Early Cretaceous and older directions lie far distant from the mean direction of the Cutler.

A major uncertainty in determining a mineralization age by this comparison is the amount of tectonic correction applied to the in situ mean direction of the Cutler; the time of tilting relative to acquisition of remanence is not independently constrained. Throughout the range of tectonic correction, the Cutler mean direction lies close to the path of Late Cretaceous and Tertiary average directions; greatest disparity is between the Cutler and late Tertiary directions. In a broad sense, then, this comparison indicates a Late Cretaceous to middle Tertiary age for uranium mineralization. Closest agreement between the Cutler mean direction and average North American directions occurs after applying the maximum tilt correction of 10°. With this correction, the Cutler mean direction (D = 358.1°, I = 65.5°; Table 1) intersects the confidence cones of the 70- and 60-m.y. directions (Fig. 10). If this 10° correction is valid, the data strongly suggest a Late Cretaceous to early Tertiary time of mineralization.

An additional difficulty in tightly defining the time of uranium mineralization arises from incompletely understood factors involving the formation of the authigenic specular hematite. As discussed below, we believe that ferric hydroxide enriched in uranium and vanadium was converted to specular hematite by aging. Whether the magnetization was locked in during this conversion or was inherited from the ferric hydroxide precursors is not known. Moreover, the rate of conversion, if continuous, is difficult to assess because it depends on kinetic effects and on the types and grain sizes (themselves unknown) of the precursors (see Berner, 1969; Langmuir, 1971). It is thus possible that some time elapsed between initial uranium enrichment and acquisition of stable magnetization. For this reason, and because of the uncertainty in tectonic correction, we conservatively interpret the paleomagnetic data as indicating a Late Cretaceous to middle Tertiary age of mineralization.

ORIGIN OF URANIUM DEPOSITS IN THE CUTLER FORMATION

The approximate mineralization age and petrologic observations constrain the source of uranium, the events and factors responsible for its introduction into the Cutler, and the geochemical conditions under which it was mobilized, transported, and concentrated. These conditions are pertinent not only to uranium mineralization but also to authigenesis of specular hematite and thus to factors responsible for chemical remagnetization of the Cutler channel sandstones.

An intrinsic source of uranium involving redistribution from within the Cutler Formation to form the present orebodies is discounted primarily because, except for the oxidized deposits in Lisbon Valley, the Cutler lacks significant uranium concentrations. This unit, unlike the Chinle and Morrison Formations, apparently never contained reducing environments of the type necessary to trap large quantities of uranium. For this reason, and because of the limited extent of the Cutler ore, an extrinsic uranium source is required. A source from the Chinle is strongly favored because of the proximity of the Cutler to Chinle deposits and because of similarities, in addition to uranium, in the geochemical signatures of their orebodies. For example, we suspect that abundant vanadium- and copper-bearing minerals characteristic of Chinle deposits in the Lisbon Valley area (Weir and Puffett, 1981; Schmitt, 1968; Gross, 1956) were the sources for the oxidized vanadium and copper phases in the Cutler deposits.

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The paleomagnetic and petrologic data and information on the structural development of the Lisbon Valley region strongly suggest a causal link between the growth of the Lisbon Valley anticline during the Late Cretaceous to early or middle Tertiary and uranium mineralization. We propose that anticlinal uplift of the Lisbon Valley area led to Cutler mineralization in the following steps. First, Chinle deposits, inferred to have existed near the axis of the Lisbon Valley anticline and updip from the present Cutler orebodies, were uplifted to elevations (perhaps above the water table) where they were subjected to conditions promoting orebody destruction. There is no direct evidence that these Chinle deposits existed. The inference for former distribution of these deposits updip from the Cutler is based on the observation that numerous deposits occur in a confined zone (Big Indian mineral belt) at the base of the Chinle Formation and probably were more widespread before erosion partly removed the ore-bearing horizon in the Chinle (Figs. 2 and 11).

The second step involved mobilization of elemental components (U, V, Cu, and Fe) from the destroyed Chinle ores and their transport in solution into Cutler sandstone. The process of mobilization and transport of Chinle uranium and vanadium must have occurred in ground-water solutions without intervening erosion, which would have dispersed these elements. Iron released from Fe-S and Fe-Cu-S minerals became the major source for ore-stage specular hematite in the Cutler ores. Minor amounts of iron were probably derived from partly dissolved ferric oxide minerals within the Cutler. Access to the Cutler channel sandstones may have been achieved by a combination of two factors. The hypothesized redistribution of uranium from the Chinle to the Cutler requires a close spatial relationship between the parent orebodies and the present ones. The most efficient redistribution would have occurred in strata where large Chinle deposits lay directly on coarse fluvial sandstones in the Cutler. This relationship is schematically illustrated in Figure 11. Removal of calcite cement, which is inferred from petrographic evidence to have been more abundant than at present, was also required to increase transmissivity of the Cutler host beds and thereby to allow ingress of mineralizing solutions. Reasons for this and other alterations, including dissolution of solid-phase ferric oxide (specular and pigmentary hematite), and mobilization, transport, and precipitation of iron and of uranium species are now addressed.

Because pH and Eh are important controls on solubility and speciation of iron and uranium at low temperatures (Langmuir, 1978; Giblin and others, 1981), these two variables are emphasized in the following paragraphs. Here, we show that an oxidizing solution of initially low pH, which evolves to one of higher pH, can explain simultaneous transport of dissolved iron and uranium and the subsequent precipitation of uranium-enriched specular hematite. Evidence for an initially acidic solution stems from the conclusion that before mineralization calcite and hematite were partly to extensively dissolved in the Cutler host beds. The probable mechanism for producing such an acidic fluid is the oxidation of sulfide minerals, which are abundant in the Chinle ores of the Big Indian mineral belt. This mechanism involves oxygen consumption by FeS_2 minerals and consequent production of hydrogen ions (equation 1; Goldhaber and others, 1978).

$$2FeS_2 + 7\tfrac{1}{2}O_2 + 5H_2O \rightarrow 2FeO(OH) + 8H^+ + 4SO_4^{2-}. \quad (1)$$

This net reaction is the result of a sequence of intermediate steps (Garrels and Thompson, 1960; Barnes and Romberger, 1968; Singer and

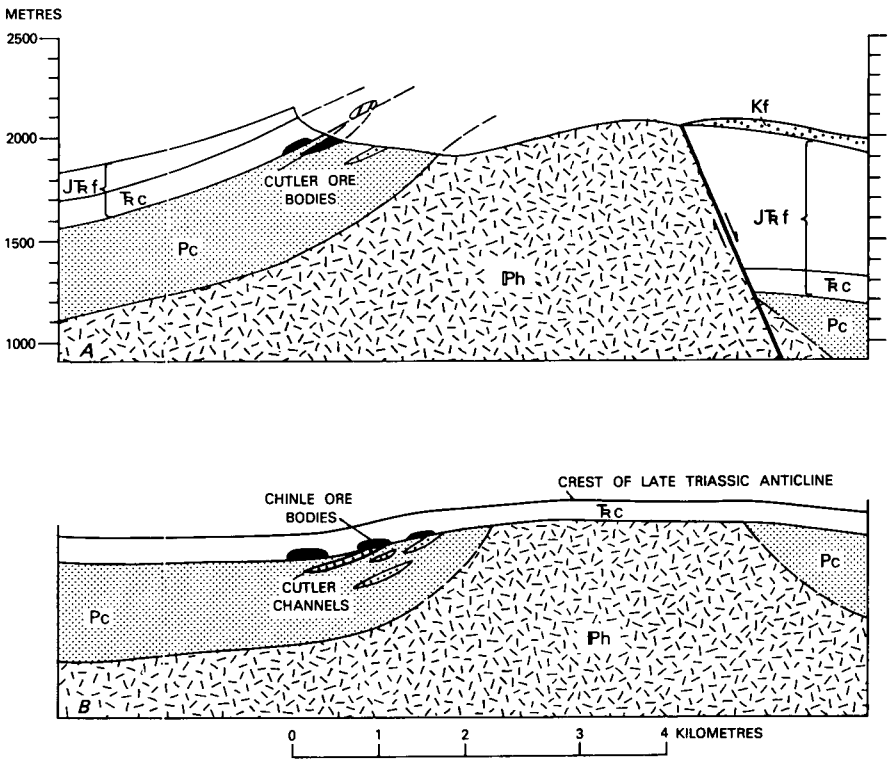


Figure 11. Schematic cross sections of the Lisbon Valley area, Utah. Section A shows present geologic relations and postulated redistribution of uranium from the Chinle Formation (TR c) into the Cutler Formation (Pc). Section B shows hypothetical relations during Late Triassic time. Other symbols as in Figure 2. Modified from Butler and Fischer (1978).

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Stumm, 1970; Nordstrom, 1982; Lowson, 1982). The initial reaction (equation 2) involves oxidation of pyrite to give ferrous iron plus sulfate and leads to an initial drop in pH of the solution:

$$2FeS_2 + 2H_2O + 7O_2 \rightarrow 2Fe^{2+} + 4H^+ + 4SO_4^{2-}. \quad (2)$$

Ferrous iron is then oxidized to ferric iron in a step (equation 3) which is strongly susceptible to bacterial catalysis:

$$2Fe^{2+} + 2H^+ + \tfrac{1}{2}O_2 \rightarrow 2Fe^{3+} + H_2O. \quad (3)$$

Ferric iron is a very effective oxidant for pyrite and reacts much more rapidly than oxygen to further depress the pH (equation 4):

$$14Fe^{3+} + FeS_2 + 8H_2O \rightarrow 15Fe^{2+} + 2SO_4^{2-} + 16H^+. \quad (4)$$

The solubility of iron oxyhydroxides increases markedly below pH 4, and eventually the sequence of reactions (equations 2–4) can produce strongly acidic solutions capable of dissolving such hydroxide phases. In this situation, appreciable ferric iron will remain in solution (equations 2–4) rather than precipitating as a solid phase (equation 1). In addition, uranium (IV) minerals (for example, uraninite—UO_2) undergoing oxidation at low pH will yield uranyl ion (UO_2^{2+}), as shown in equation 5:

$$UO_2 + \tfrac{1}{2}O_2 + 2H^+ \rightarrow UO_2^{2+} + H_2O. \quad (5)$$

Thus, UO_2^{2+} and Fe^{3+} may be released from the site of orebody destruction.

Acidic (pH 4–5) solutions are effective media for simultaneously transporting major concentrations of iron, as aqueous Fe^{3+}, and uranium, as UO_2^{2+} (see Langmuir, 1978; Giblin and others, 1981). These low pH values are considered reasonable in view of the extreme acidity (pH <3) of acid mine waters that results from identical processes (Barnes and Romberger, 1968; Singer and Stumm, 1970). Given probable acidic conditions, co-mobility of Fe^{3+} and UO_2^{2+} additionally requires an oxidizing fluid (Eh >0). Similarly, acidic oxidizing solutions are capable of transporting cationic vanadium (Hostetler and Garrels, 1962) and copper (Rose, 1976).

Concentration of iron and uranium to form the Cutler ores was most likely related to changes in the pH of the mineralizing solution upon contact with mineral constituents within the channel sandstones. We hypothesize that acidic waters, which were responsible for transport of UO_2^{2+} and Fe^{3+} and for the dissolution of calcite cement and hematite (martite and pigment) in the Cutler host beds, were neutralized primarily by reactions with the calcite and perhaps with authigenic clays. Increase in pH via this mechanism would cause precipitation of ferric hydroxide from mobile ferric iron (equation 6):

$$H^+ + Fe^{3+} + 4CaCO_{3(s)} + 3H_2O \rightarrow Fe(OH)_3 + 4Ca^{2+} + 4HCO_3^-. \quad (6)$$

Precipitation of initially hydrous ferric oxides provided a substrate with a high sorptive capacity for dissolved uranium and vanadium (Van der Weijden and others, 1976; Barton, 1956; Muto and others, 1968; Lovering, 1955; Langmuir, 1978; Zielinski and others, 1982). Importantly, adsorption of uranyl onto ferric oxyhydroxide is pH dependent: negligible adsorption occurs at low pH (<5), whereas maximum adsorption is achieved at pH values above 5–6 (Langmuir, 1978; Hsi and Langmuir, 1980). Neutralization of acidic mineralizing fluids thus promotes both precipitation of ferric hydroxide phases and adsorption of uranyl onto these oxides. Finally, specular hematite formed by aging of thermodynamically metastable ferric hydroxides (Berner, 1969; Langmuir, 1971), and during this process, at least some of its adsorbed uranium and vanadium was retained. Uranium and vanadium released during formation of hematite (Van der Weijden and others, 1976) probably entered associated secondary minerals as carnotite and bequerelite.

CONCLUSIONS

Uranium ore deposits in the Lower Permian Cutler Formation occur in purple and red, coarse-grained fluvial sandstones on the southwest flank of the Lisbon Valley anticline, Paradox Basin, Utah. The Cutler deposits are characterized by ore minerals in which uranium and vanadium are in oxidized valence states, by abundant authigenic specular hematite enriched in uranium and vanadium, and by close spatial relationship to the overlying Triassic Chinle Formation. In the Lisbon Valley area, the Chinle contains numerous uranium deposits, but these occur in geochemically reduced environments.

Paleomagnetic studies, based on comparing remanent directions in ore-bearing sandstones to those in nearby unmineralized sandstone and to mean directions for stable North America at 10-m.y. intervals since the Permian, yield an approximate age of uranium mineralization of the Cutler orebodies. Thermal cleaning of mineralized sandstone yields a tilt-corrected mean direction close to predicted Late Cretaceous to middle Tertiary directions at Lisbon Valley. The stable remanence in mineralized rock is carried by authigenic uranium-bearing specular hematite, and it thereby records the time of mineralization. The contribution of Permian magnetic components in mineralized rock has been almost entirely eliminated by partial dissolution of martite before specular hematite authigenesis. These results, combined with information on the structural development of the Lisbon Valley area and on nearby uranium deposits in the Triassic Chinle Formation, suggest that uplift during the Late Cretaceous to middle Tertiary promoted destruction of Chinle orebodies and led to redistribution of iron and uranium from the Chinle into the Cutler. Acidic solutions may account for the transport of significant quantities of iron and uranium species as well as for the partial dissolution of martite in the Cutler. Neutralization of these solutions by reaction with calcite caused precipitation of amorphous ferric oxide. Ferric oxide was converted to specular hematite which had retained uranium by adsorption.

ACKNOWLEDGMENTS

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